

Data redundancy and inc... credit_card_file x

Limit to 1000 r

```
1 • SELECT * FROM bad_bankingdb.credit_card_file;
```

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content:

	card_id	customer_id	customer_name	address	phone	card_number	card_type	credit_limit
▶	1	C001	Aung Aung	No.10, Pyay Road, Yangon	09-111111111	4111111111111111	Gold	2000000.00
	2	C003	Kyaw Kyaw	No.45, Mandalay Road, Mandalay	09-333333333	4222222222222222	Classic	1000000.00
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Data redundancy and inc... x

Limit to 1000 r

```
88 • UPDATE account_file
89 SET address = 'No.99, Inya Road, Yangon'
90 WHERE customer_id = 'C001';
91
92 -- Check inconsistency
93
94 • SELECT 'ACCOUNT FILE' AS source, customer_id, customer_name, address
95 FROM account_file
96 WHERE customer_id = 'C001'
97 UNION ALL
98 SELECT 'LOAN FILE' AS source, customer_id, customer_name, address
99 FROM loan_file
100 WHERE customer_id = 'C001'
101 UNION ALL
102 SELECT 'CREDIT CARD FILE' AS source, customer_id, customer_name, address
103 FROM credit_card_file
104 WHERE customer_id = 'C001';
105
```

Result Grid Filter Rows: Export: Wrap Cell Content:

	source	customer_id	customer_name	address
▶	ACCOUNT FILE	C001	Aung Aung	No.99, Inya Road, Yangon
	LOAN FILE	C001	Aung Aung	No.10, Pyay Road, Yangon
	CREDIT CARD FILE	C001	Aung Aung	No.10, Pyay Road, Yangon

Data redundancy and inc...

customers

Limit to 1000 r

1 •

SELECT * FROM good_bankingdb.customers;

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

	customer_id	customer_name	address	phone	email	created_at
▶	C001	Aung Aung	No.10, Pyay Road, Yangon	09-111111111	aung@example.com	2026-04-18 21:15:03
	C002	Su Su	No.22, Hledan Street, Yangon	09-222222222	susu@example.com	2026-04-18 21:15:03
	C003	Kyaw Kyaw	No.45, Mandalay Road, Mandalay	09-333333333	kyaw@example.com	2026-04-18 21:15:03
*	NULL	NULL	NULL	NULL	NULL	NULL

Data redundancy and inc...

accounts

Limit to 1000 r

1 •

SELECT * FROM good_bankingdb.accounts;

Result Grid

Filter Rows:

Edit:

Export/Import:

	account_id	customer_id	account_type	balance	opened_date
▶	1	C001	Savings	500000.00	2024-01-10
	2	C002	Current	1200000.00	2024-02-15
	3	C003	Savings	300000.00	2024-03-01
*	NULL	NULL	NULL	NULL	NULL

Data redundancy and inc... loans x

Limit to 1000 r

1 • `SELECT * FROM good_bankngdb.loans;`

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Cont

	loan_id	customer_id	loan_type	loan_amount	outstanding_amount	start_date
▶	1	C001	Home Loan	10000000.00	7500000.00	2024-04-01
	2	C002	Car Loan	5000000.00	2000000.00	2024-05-10
*	NULL	NULL	NULL	NULL	NULL	NULL

Data redundancy and inc... credit_cards x

Limit to 1000 r

1 • `SELECT * FROM good_bankngdb.credit_cards;`

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell C

	card_id	customer_id	card_number	card_type	credit_limit	expiry_date
▶	1	C001	4111111111111111	Gold	2000000.00	2028-12-31
	2	C003	4222222222222222	Classic	1000000.00	2027-11-30
*	NULL	NULL	NULL	NULL	NULL	NULL

Data redundancy and inc... x

Limit to 1000 r

```

207 • UPDATE customers
208   SET address = 'No.99, Inya Road, Yangon'
209  WHERE customer_id = 'C001';
210
211  -- Check consistent result through joins
212 • SELECT
213     c.customer_id,
214     c.customer_name,
215     c.address,
216     a.account_type,
217     a.balance,
218     l.loan_type,
219     l.outstanding_amount,
220     cc.card_number,
221     cc.card_type
222  FROM customers c
223 LEFT JOIN accounts a ON c.customer_id = a.customer_id
224 LEFT JOIN loans l ON c.customer_id = l.customer_id
225 LEFT JOIN credit_cards cc ON c.customer_id = cc.customer_id
226  WHERE c.customer_id = 'C001';
227

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: I

	customer_id	customer_name	address	account_type	balance	loan_type	outstanding_amount	card_number	card_type
▶	C001	Aung Aung	No.99, Inya Road, Yangon	Savings	500000.00	Home Loan	7500000.00	4111111111111111	Gold

Data redundancy and inc... x

Limit to 1000 r

```

239     a.balance,
240     l.loan_id,
241     l.loan_type,
242     l.loan_amount,
243     l.outstanding_amount,
244     cc.card_id,
245     cc.card_number,
246     cc.card_type,
247     cc.credit_limit
248  FROM customers c
249 LEFT JOIN accounts a ON c.customer_id = a.customer_id
250 LEFT JOIN loans l ON c.customer_id = l.customer_id
251 LEFT JOIN credit_cards cc ON c.customer_id = cc.customer_id;
252
253 • SELECT * FROM vw_customer_banking_profile WHERE customer_id = 'C001';
254

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: I

	customer_id	customer_name	address	phone	account_id	account_type	balance	loan_id	loan_type	loan_amount	outstanding_amount	card_id	card_number	card_type	credit_limi
▶	C001	Aung Aung	No.99, Inya Road, Yangon	09-111111111	1	Savings	500000.00	1	Home Loan	10000000.00	7500000.00	1	4111111111111111	Gold	2000000.0

Data redundancy and inc...

Limit to 1000 r

```

272  -- Update only account file
273 •  UPDATE account_file
274     SET address = 'No.99, Inya Road, Yangon'
275     WHERE customer_id = 'C001';
276
277  -- Verify inconsistency
278 •  SELECT 'ACCOUNT FILE' AS source, address
279     FROM account_file
280     WHERE customer_id = 'C001'
281     UNION ALL
282     SELECT 'LOAN FILE' AS source, address
283     FROM loan_file
284     WHERE customer_id = 'C001'
285     UNION ALL
286     SELECT 'CREDIT CARD FILE' AS source, address
287     FROM credit_card_file
288     WHERE customer_id = 'C001';

```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	source	address
▶	ACCOUNT FILE	No.99, Inya Road, Yangon
	LOAN FILE	No.10, Pyay Road, Yangon
	CREDIT CARD FILE	No.10, Pyay Road, Yangon

```
298 • UPDATE customers
299 SET address = 'No.99, Inya Road, Yangon'
300 WHERE customer_id = 'C001';
301
302 -- Verify consistency
303 • SELECT
304     c.customer_id,
305     c.customer_name,
306     c.address,
307     a.account_type,
308     l.loan_type,
309     cc.card_type
310 FROM customers c
311 LEFT JOIN accounts a ON c.customer_id = a.customer_id
312 LEFT JOIN loans l ON c.customer_id = l.customer_id
313 LEFT JOIN credit_cards cc ON c.customer_id = cc.customer_id
314 WHERE c.customer_id = 'C001';
```

Result Grid  Filter Rows: Export:  Wrap Cell Content: 

	customer_id	customer_name	address	account_type	loan_type	card_type
▶	C001	Aung Aung	No.99, Inya Road, Yangon	Savings	Home Loan	Gold